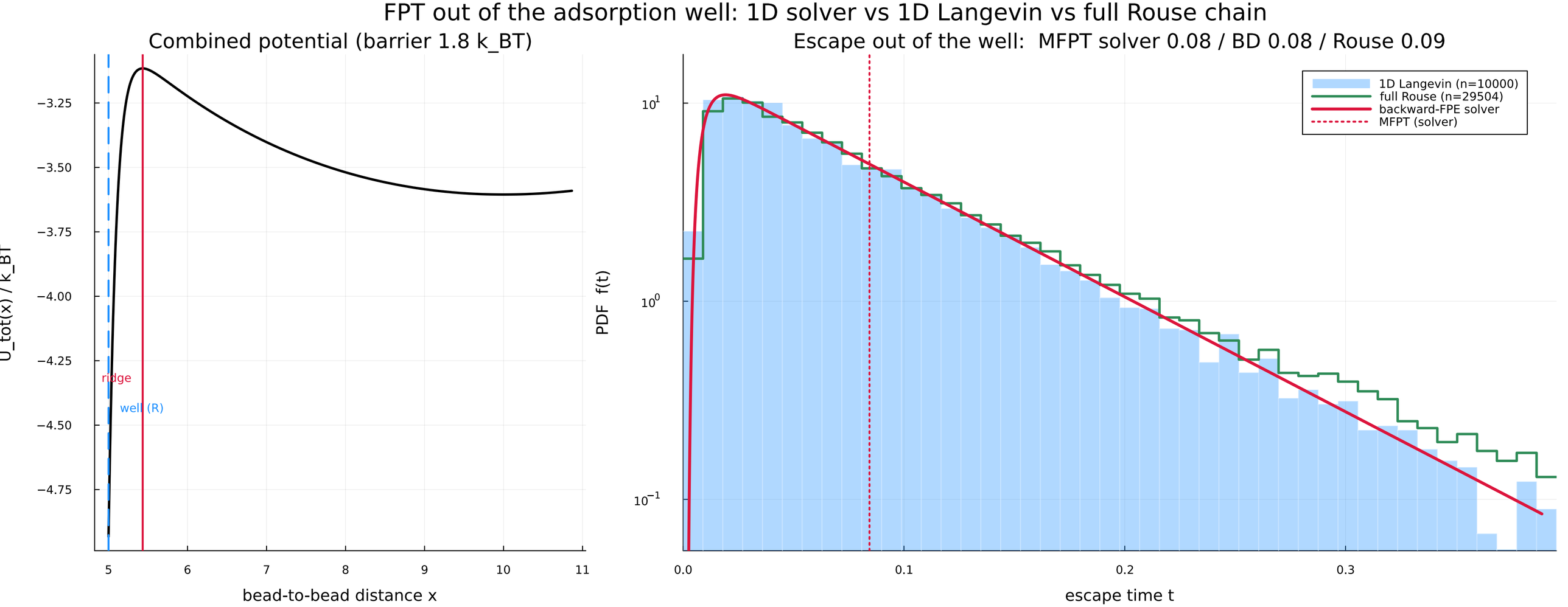
FPT out of the adsorption well: 1D solver vs 1D Langevin vs full Rouse chain
Combined potential (barrier 1.8 k_BT)
U_tot(x) / k_BT
−3.25
−3.50
−3.75
−4.00
−4.25
−4.50
−4.75
ridge
well (R)
5
6
7
8
9
10
11
bead-to-bead distance x
Escape out of the well: MFPT solver 0.08 / BD 0.08 / Rouse 0.09
PDF f(t)
10^{1}
10^{0}
10^{-1}
1D Langevin (n=10000)
full Rouse (n=29504)
backward-FPE solver
MFPT (solver)
0.0
0.1
0.2
0.3
escape time t